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Applied Calculus IV

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Week 4

- Review
- 2.4
- 2.5

Contents

Preface vii



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- 1 Introduction to Differential Equations 2**
 - 1.1 Definitions and Terminology 3
 - 1.2 Initial-Value Problems 15
 - 1.3 Differential Equations as Mathematical Models 22
 - CHAPTER 1 IN REVIEW 34**



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- 2 First-Order Differential Equations 36**
 - 2.1 Solution Curves Without a Solution 37
 - 2.1.1 Direction Fields 37
 - 2.1.2 Autonomous First-Order DEs 39
 - 2.2 Separable Equations 47
 - 2.3 Linear Equations 55
 - 2.4 Exact Equations 64
 - 2.5 Solutions by Substitutions 72
 - 2.6 A Numerical Method 76
 - CHAPTER 2 IN REVIEW 81**



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- 3 Modeling with First-Order Differential Equations 84**
 - 3.1 Linear Models 85
 - 3.2 Nonlinear Models 96
 - 3.3 Modeling with Systems of First-Order DEs 107
 - CHAPTER 3 IN REVIEW 114**



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- 4 Higher-Order Differential Equations 118**
 - 4.1 Preliminary Theory—Linear Equations 119
 - 4.1.1 Initial-Value and Boundary-Value Problems 119
 - 4.1.2 Homogeneous Equations 121
 - 4.1.3 Nonhomogeneous Equations 127
 - 4.2 Reduction of Order 132
 - 4.3 Homogeneous Linear Equations with Constant Coefficients 135
 - 4.4 Undetermined Coefficients—Superposition Approach 142
 - 4.5 Undetermined Coefficients—Annihilator Approach 152
 - 4.6 Variation of Parameters 159
 - 4.7 Cauchy-Euler Equations 166

First-Order Differential Equations

- 2.1 Solution Curves Without a Solution
- 2.2 Separable Equations
- 2.3 Linear Equations
- 2.4 Exact Equations
- 2.5 Solutions by Substitutions
- 2.6 A Numerical Method

Recall

Differential of $f(x, y)$

$$z = f(x, y) \rightarrow dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

$$c = f(x, y) \rightarrow 0 = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

(a function of two variables with continuous first partial derivatives in a region R of the xy -plane)

Example:

$$x^2 + 2xy + y^3 = c \quad \rightarrow \quad (2x + 2y)dx + (2x + 3y^2)dy = 0$$

$$e^x - y + 2xy = c \quad \rightarrow \quad (e^x + 2y)dx + (2x - 1)dy = 0$$

$$x^3 - ye^x + 2x \sin y = c$$

$$\rightarrow (3x^2 - ye^x + 2 \sin y)dx + (-e^x + 2x \cos y)dy = 0$$

Question: Can we do it reversely?

(can we find the solution of a differential equation)

Solution

DE

$$x^2 + 2xy + y^3 = c \quad \leftarrow \quad (2x + 2y)dx + (2x + 3y^2)dy = 0$$

$$e^x - y + 2xy = c \quad \leftarrow \quad (e^x + 2y)dx + (2x - 1)dy = 0$$

Let us write our DE in this form : $M(x, y)dx + N(x, y)dy = 0$

If we could find a function $f(x, y)$ such that $M(x, y) = \frac{\partial f}{\partial x}$ & $N(x, y) = \frac{\partial f}{\partial y}$

obviously, $f(x, y) = c$ is the solution of the DE.

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

Note that **not** every first-order DE written in differential form $M(x, y) dx + N(x, y) dy = 0$ corresponds to a differential of $f(x, y) = c$.

DEFINITION 2.4.1 Exact Equation

A differential expression $M(x, y) dx + N(x, y) dy$ is an **exact differential** in a region R of the xy -plane if it corresponds to the differential of some function $f(x, y)$ defined in R . A first-order differential equation of the form

$$M(x, y) dx + N(x, y) dy = 0$$

is said to be an **exact equation** if the expression on the left-hand side is an exact differential.

* **Not** every first-order DE is an exact differential equation.

Example

$$(2x + 2y)dx + (2x + 3y^2)dy = 0$$

$$\frac{\partial f}{\partial x} = 2x + 2y \quad \& \quad \frac{\partial f}{\partial y} = 2x + 3y^2$$

$$\Rightarrow f(x, y) = x^2 + 2xy + y^3$$

Since we could find $f(x, y) = c$, this is an exact differential equation

Recall

If f_x and f_y are continuous, f_{xy} and f_{yx} exist, and are continuous in a rectangular region R

$$f_{xy} = f_{yx}$$

or
$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

From the other hand,

$$M(x, y)dx + N(x, y)dy = 0$$


$$\frac{\partial f}{\partial x}$$


$$\frac{\partial f}{\partial y}$$



$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

THEOREM 2.4.1 Criterion for an Exact Differential

Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous first partial derivatives in a rectangular region R defined by $a < x < b, c < y < d$. Then a necessary and sufficient condition that $M(x, y) dx + N(x, y) dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \quad (4)$$

Method of Solution?

Assume it is an exact DE

$$M_y = N_x$$

$$M(x, y)dx + N(x, y)dy = 0$$



$$\frac{\partial f}{\partial x}$$



$$\frac{\partial f}{\partial y}$$

$$f(x, y) = \int M(x, y)dx + g(y)$$

$$\Rightarrow \frac{\partial f(x, y)}{\partial y} = \frac{\partial}{\partial y} \int M(x, y)dx + \frac{\partial}{\partial y} g(y) = N(x, y) \Rightarrow \frac{\partial}{\partial y} g(y) = N(x, y) - \frac{\partial}{\partial y} \int M(x, y)dx$$

$$g(y) = \int \left(N(x, y) - \frac{\partial}{\partial y} \int M(x, y)dx \right) dy$$

Alternatively,

$$M(x, y)dx + N(x, y)dy = 0$$



$$\frac{\partial f}{\partial x}$$



$$\frac{\partial f}{\partial y}$$

$$f(x, y) = \int N(x, y)dy + h(x)$$

$$\Rightarrow \frac{\partial f(x, y)}{\partial x} = \frac{\partial}{\partial x} \int N(x, y)dy + \frac{\partial}{\partial x} h(x) = M(x, y) \Rightarrow \frac{\partial}{\partial x} h(x) = M(x, y) - \frac{\partial}{\partial x} \int N(x, y)dy$$

$$h(x) = \int \left(M(x, y) - \frac{\partial}{\partial x} \int N(x, y)dy \right) dx$$

Summary

If $M_y = N_x$

Step 1.
$$f(x, y) = \int M(x, y) dx + g(y)$$

Step 2.
$$g(y) = \int \left(N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx \right) dy$$

Step 1.
$$f(x, y) = \int N(x, y) dy + h(x)$$

Step 2.
$$h(x) = \int \left(M(x, y) - \frac{\partial}{\partial x} \int N(x, y) dy \right) dx$$

Depending on the difficulty of integrations, you can choose one way

Example: Slove $2xydx + (x^2 - 1)dy = 0$

$\downarrow$ $\downarrow$
 $M(x, y)$ $N(x, y)$

$$M_y = N_x = 2x \quad \checkmark$$

$$f(x, y) = \int M(x, y)dx + g(y) = \int 2xydx + g(y) = yx^2 + g(y)$$

$$\frac{\partial f(x, y)}{\partial y} = x^2 + \frac{d}{dy}g(y) = N(x, y) = x^2 - 1 \Rightarrow \frac{\partial}{\partial y}g(y) = x^2 - 1 - x^2$$

$$\Rightarrow g(y) = -y$$

$$\Rightarrow f(x, y) = yx^2 - y$$

$$\Rightarrow yx^2 - y = c \Rightarrow y = \frac{c}{x^2 - 1} \quad \text{defined on } (1, \infty)$$

Example: Slove $\frac{dy}{dx} = -\frac{2xy}{x^2+3y^2+1}$ $\longrightarrow$ $2xydx + (x^2 + 3y^2 + 1)dy = 0$

$\downarrow$ $\downarrow$
 $M(x, y)$ $N(x, y)$

$$M_y = N_x = 2x \quad \checkmark$$

$$f(x, y) = \int M(x, y)dx + g(y) = \int 2xydx + g(y) = yx^2 + g(y)$$

$$\frac{\partial f(x, y)}{\partial y} = x^2 + \frac{d}{dy}g(y) = N(x, y) = x^2 + 3y^2 + 1 \Rightarrow \frac{\partial}{\partial y}g(y) = x^2 + 3y^2 + 1 - x^2$$

$$\Rightarrow g(y) = y^3 + y$$

$$\Rightarrow f(x, y) = yx^2 + y^3 + y$$

$$\Rightarrow y(x^2+1) + y^3 = c$$

Example: Slove $\frac{dy}{dx} = -\frac{2xy}{x^2+3y^2+1}$ $\longrightarrow$ $2xydx + (x^2 + 3y^2 + 1)dy = 0$

$\downarrow$ $\downarrow$
 $M(x, y)$ $N(x, y)$

$$M_y = N_x = 2x \quad \checkmark$$

$$f(x, y) = \int M(x, y)dx + g(y) = \int 2xydx + g(y) = yx^2 + g(y)$$

$$\frac{\partial f(x, y)}{\partial y} = x^2 + \frac{\partial}{\partial y}g(y) = N(x, y) = x^2 + 3y^2 + 1 \Rightarrow \frac{\partial}{\partial y}g(y) = x^2 + 3y^2 + 1 - x^2$$

$$\Rightarrow g(y) = y^3 + y$$

$$\Rightarrow f(x, y) = yx^2 + y^3 + y$$

$$\Rightarrow y(x^2+1) + y^3 = c$$

Exercises 2.4

In Problems 1–20 determine whether the given differential equation is exact. If it is exact, solve it.

→ 1. $(2x - 1) dx + (3y + 7) dy = 0$

2. $(2x + y) dx - (x + 6y) dy = 0$

3. $(5x + 4y) dx + (4x - 8y^3) dy = 0$

→ 4. $(\sin y - y \sin x) dx + (\cos x + x \cos y - y) dy = 0$

5. $(2xy^2 - 3) dx + (2x^2y + 4) dy = 0$

6. $\left(2y - \frac{1}{x} + \cos 3x\right) \frac{dy}{dx} + \frac{y}{x^2} - 4x^3 + 3y \sin 3x = 0$

7. $(x^2 - y^2) dx + (x^2 - 2xy) dy = 0$

8. $\left(1 + \ln x + \frac{y}{x}\right) dx = (1 - \ln x) dy$

9. $(x - y^3 + y^2 \sin x) dx = (3xy^2 + 2y \cos x) dy$

10. $(x^3 + y^3) dx + 3xy^2 dy = 0$

1. Solve $(2x - 1)dy + (3y + 7)dx = 0$

$$M(x, y)$$

$$N(x, y)$$

$$M_y = N_x = 0 \quad \checkmark$$

$$f(x, y) = \int (2x - 1)dx + g(y) = x^2 - x + g(y)$$

$$\frac{\partial f(x, y)}{\partial y} = g'(y) = N(x, y) = 3y + 7$$

$$\Rightarrow g' = 3y + 7 \quad \Rightarrow g(y) = \frac{3}{2}y^2 + 7y$$

$$x^2 - x + \frac{3}{2}y^2 + 7y = c$$

$$4. (\sin y - y \sin x) dx + (\cos x + x \cos y - y) dy = 0$$

$$M(x, y)$$

$$N(x, y)$$

$$M_y = N_x = \cos y - \sin x \quad \checkmark$$

$$f(x, y) = \int (\sin y - y \sin x) dx + g(y) = x \sin y + y \cos x + g(y)$$

$$\frac{\partial f(x, y)}{\partial y} = x \cos y + \cos x + g'(y) = N(x, y) = \cos x + x \cos y - y$$

$$\Rightarrow g' = \cos x + x \cos y - y - x \cos y + \cos x = -y$$

$$\Rightarrow g(y) = -\frac{1}{2}y^2$$

$$\Rightarrow x \sin y + y \cos x - \frac{1}{2}y^2 = c$$

7. Solve $(x^2 - y^2)dy + (x^2 - 2xy)dx = 0$

$$M(x, y) \quad N(x, y)$$

$$M_y \neq N_x$$

The equation is not an exact differential equation

In Problems 21–26 solve the given initial-value problem.

21. $(x + y)^2 dx + (2xy + x^2 - 1) dy = 0, \quad y(1) = 1$

22. $(e^x + y) dx + (2 + x + ye^y) dy = 0, \quad y(0) = 1$

23. $(4y + 2t - 5) dt + (6y + 4t - 1) dy = 0, \quad y(-1) = 2$

24. $\left(\frac{3y^2 - t^2}{y^5}\right) \frac{dy}{dt} + \frac{t}{2y^4} = 0, \quad y(1) = 1$

25. $(y^2 \cos x - 3x^2y - 2x) dx$
 $+ (2y \sin x - x^3 + \ln y) dy = 0, \quad y(0) = e$

26. $\left(\frac{1}{1 + y^2} + \cos x - 2xy\right) \frac{dy}{dx} = y(y + \sin x), \quad y(0) = 1$

Similar to the previous exercises (and at the end, find the value of parameter c by using the initial condition)

Challenge: What if $M_y \neq N_x$?

We search for an Integrating Factors

$$\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0$$

If it becomes an exact differential, then

$$\mu_y M + \mu M_y = \mu_x N + \mu N_x$$

$$\mu(M_y - N_x) = \mu_x N - \mu_y M$$

if $\mu_x = 0$

$$\frac{d\mu}{dy} = -\frac{M_y - N_x}{M}\mu$$

if $\mu_y = 0$

$$\frac{d\mu}{dx} = \frac{M_y - N_x}{N}\mu$$

Conclusion (integrating factor for non-exact DE)

If $\frac{N_x - M_y}{M}$ is a function of y alone, $\mu(y) = e^{\int \frac{N_x - M_y}{M} dy}$

Why?

If $\frac{M_y - N_x}{N}$ is a function of x alone, $\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}$

Why?

Example: Slove $xydx + (2x^2 + 3y^2 - 20)dy = 0$

$$M_y \neq N_x \quad M_y - N_x = x - 4x = -3x$$

$$\frac{M_y - N_x}{N} = \frac{-3x}{2x^2 + 3y^2 - 20} \text{ is not a function of } x \text{ alone}$$

$$\frac{N_x - M_y}{M} = \frac{3x}{xy} = \frac{3}{y} \text{ is a function of } y \text{ alone} \Rightarrow \mu(y) = e^{\int \frac{3}{y}} = y^3 \quad \text{Integrating Factor}$$

$$\Rightarrow xy^4dx + (2x^2y^3 + 3y^5 - 20y^3)dy = 0$$

Now $xy^4dx + (2x^2y^3 + 3y^5 - 20y^3)dy = 0$ is an exact DE

$$M(x, y)$$

$$N(x, y)$$

$$M_y = N_x = 4xy^3 \quad \checkmark$$

$$f(x, y) = \int M(x, y)dx + g(y) = \int xy^4dx + g(y) = \frac{1}{2}x^2y^4 + g(y)$$

$$\frac{\partial f(x, y)}{\partial y} = 2x^2y^3 + g' = N(x, y) = 2x^2y^3 + 3y^5 - 20y^3 \Rightarrow g' = 2x^2y^3 + 3y^5 - 20y^3 - 2x^2y^3$$

$$\Rightarrow g(y) = \frac{3}{6}y^6 - 5y^4$$

$$\Rightarrow f(x, y) = \frac{1}{2}x^2y^4 + \frac{3}{6}y^6 - 5y^4$$

$$\Rightarrow \frac{1}{2}x^2y^4 + \frac{1}{3}y^6 - 5y^4 = c$$

In Problems 27 and 28 find the value of k so that the given differential equation is exact.

27. $(y^3 + kxy^4 - 2x) dx + (3xy^2 + 20x^2y^3) dy = 0$

28. $(6xy^3 + \cos y) dx + (2kx^2y^2 - x \sin y) dy = 0$

$$M_y = N_x$$

27.

$$M_y = 3y^2 + 4kxy^3$$

$$N_x = 3y^2 + 40xy^3$$

$$3y^2 + 4kxy^3 = 3y^2 + 40xy^3$$

$$k = 10$$

28.

$$M_y = 18xy^2 - \sin y$$

$$N_x = 4kxy^2 - \sin y$$

$$18xy^2 - \sin y = 4kxy^2 - \sin y$$

$$k = \frac{9}{2}$$

In Problems 31–36 solve the given differential equation by finding, as in Example 4, an appropriate integrating factor.

31. $(2y^2 + 3x) dx + 2xy dy = 0$

32. $y(x + y + 1) dx + (x + 2y) dy = 0$

33. $6xy dx + (4y + 9x^2) dy = 0$

34. $\cos x dx + \left(1 + \frac{2}{y}\right) \sin x dy = 0$

35. $(10 - 6y + e^{-3x}) dx - 2 dy = 0$

36. $(y^2 + xy^3) dx + (5y^2 - xy + y^3 \sin y) dy = 0$

$$31. (2y^2 + 3x) dx + 2xy dy = 0$$

$$M_y = 4y \text{ \& } N_x = 2y$$

$$\frac{M_y - N_x}{N} = \frac{2y}{2xy} = \frac{1}{x} \rightarrow \mu(x) = e^{\int \frac{1}{x} dx} = x \rightarrow (2xy^2 + 3x^2)dx + 2x^2ydy = 0$$

$$f_x = 2xy^2 + 3x^2$$

$$f = x^2y^2 + x^3 + h(y)$$

$$h'(y) = 0$$

$$h(y) = 0$$

$$\text{Solution: } x^2y^2 + x^3 = c$$

$$32. y(x + y + 1) dx + (x + 2y) dy = 0$$

$$M_y = x + 2y + 1 \quad \& \quad N_x = 1$$

$$\frac{M_y - N_x}{N} = \frac{x + 2y}{x + 2y} = 1 \rightarrow \mu(x) = e^{\int dx} = e^x \rightarrow y(x + y + 1)e^x dx + \underset{\substack{\downarrow \\ f_y}}{(x + 2y)e^x} dy = 0$$

$$f = xye^x + y^2e^x + h(x)$$

$$f_x = ye^x + xye^x + y^2e^x + h'(x)$$

$$h'(x) = 0$$

$$h(x) = 0$$

$$\text{Solution: } xye^x + y^2e^x = c$$

In Problems 37 and 38 solve the given initial-value problem by finding, as in Example 4, an appropriate integrating factor.

37. $x \, dx + (x^2y + 4y) \, dy = 0, \quad y(4) = 0$

38. $(x^2 + y^2 - 5) \, dx = (y + xy) \, dy, \quad y(0) = 1$

37. $x dx + (x^2y + 4y) dy = 0, \quad y(4) = 0$

$$M_y = 0 \quad \& \quad N_x = 2xy$$

$$\frac{M_y - N_x}{N} = \frac{-2xy}{x^2y + 4y} = \frac{-2x}{x^2 + 4} \rightarrow \mu(x) = e^{\int \frac{-2x}{x^2+4} dx} = e^{-\ln |x^2+4|} = \frac{1}{x^2 + 4}$$

$$\frac{x}{x^2 + 4} dx + y dy = 0 \rightarrow f = \frac{1}{2} \ln(4 + x^2) + h(y)$$

$$\downarrow$$

$$f_x$$

$$h'(y) = y$$

$$h(y) = \frac{1}{2} y^2$$

$$\text{Solution: } \frac{1}{2} \ln(4 + x^2) + \frac{1}{2} y^2 = c$$

$$\ln(4 + x^2) + y^2 = 2c$$

$$\ln(4 + x^2) + y^2 = c_1$$

$$e^{\ln(4+x^2)+y^2} = e^{c_1}$$

$$e^{\ln(4+x^2)} \cdot e^{y^2} = e^{c_1}$$

$$(4 + x^2)e^{y^2} = c_2$$

$$c_2 = 20$$

$$(4 + x^2)e^{y^2} = 20$$

37. $x dx + (x^2y + 4y) dy = 0, \quad y(4) = 0$

❖ Another integrating factor for this DE

$$M_y = 0 \quad \& \quad N_x = 2xy$$

$$\frac{N_x - M_y}{M} = \frac{2xy}{x} = 2y \rightarrow \mu(x) = e^{\int 2y dy} = e^{y^2} \rightarrow x e^{y^2} dx + y(x^2 + 4)e^{y^2} dy = 0$$

↓
 f_x

$$f = \int x e^{y^2} dx + g(y) = \frac{1}{2} x^2 e^{y^2} + g(y)$$

$$f_y = \frac{1}{2} x^2 (2y e^{y^2}) + g'(y) = y x^2 e^{y^2} + 4y e^{y^2}$$

$$\rightarrow g'(y) = 4y e^{y^2} \rightarrow g = 2e^{y^2}$$

$$f = \frac{1}{2} x^2 e^{y^2} + 2e^{y^2} \rightarrow \frac{1}{2} x^2 e^{y^2} + 2e^{y^2} = c_1 \rightarrow x^2 e^{y^2} + 4e^{y^2} = 4c_1 = c \rightarrow (x^2 + 4)e^{y^2} = c$$

$$y(4) = 0 \rightarrow c = 20$$

$$\rightarrow (x^2 + 4)e^{y^2} = 20$$

First-Order Differential Equations

- 2.1 Solution Curves Without a Solution
- 2.2 Separable Equations
- 2.3 Linear Equations
- 2.4 Exact Equations
- 2.5 Solutions by Substitutions
- 2.6 A Numerical Method

Simplify DE by substituting y
(with your guess)

Example: Slove $x \frac{dy}{dx} + y = x^2 y^2 \longrightarrow \frac{dy}{dx} + \frac{y}{x} = xy^2$

$$u = y^{-1} \rightarrow y = u^{-1} \rightarrow \frac{dy}{dx} = -u^{-2} \frac{du}{dx}$$

$$\rightarrow -u^{-2} \frac{du}{dx} + \frac{u^{-1}}{x} = xu^{-2} \rightarrow \frac{du}{dx} - \frac{u}{x} = -x$$

$$P(x) = \frac{-1}{x} \rightarrow e^{-\int \frac{1}{x} dx} = x^{-1} \rightarrow -\frac{1}{x} \frac{du}{dx} + \frac{u}{x^2} = -1 \rightarrow \frac{d}{dx} \left[\frac{u}{x} \right] = -1$$

$$\rightarrow \frac{u}{x} = -x + c \rightarrow u = -x^2 + cx \rightarrow y^{-1} = -x^2 + cx \rightarrow y = \frac{1}{-x^2 + cx}$$

Example: Slove $\frac{dy}{dx} = (-2x + y)^2 - 7, \quad y(0) = 0$

$$u = -2x + y \rightarrow \frac{du}{dx} = -2 + \frac{dy}{dx} \rightarrow \frac{du}{dx} + 2 = u^2 - 7$$

$$\frac{du}{dx} = u^2 - 9 \rightarrow \frac{du}{u^2 - 9} = dx \rightarrow \frac{1}{6} \left[\frac{1}{u-3} - \frac{1}{u+3} \right] du = dx$$

$$\frac{1}{6} \ln \frac{u-3}{u+3} = x + c \rightarrow \frac{u-3}{u+3} = e^{6x+6c} = c_1 e^x \rightarrow \frac{-2x+y-3}{-2x+y+3} = c_1 e^x \rightarrow c_1 = -1$$

$$\rightarrow \frac{-2x + y - 3}{-2x + y + 3} = -e^x$$

homogeneous function of degree α

$$f(tx, ty) = t^\alpha f(x, y)$$

Example

$$f(x, y) = x^3 + y^3$$

homogeneous function of degree 3

$$f(tx, ty) = (tx)^3 + (ty)^3 = t^3(x^3 + y^3) = t^3 f(x, y)$$

This first order differential equation is called homogeneous if M and N both are homogeneous functions of the same degree

$$M(x, y) dx + N(x, y) dy = 0$$

In Problems 1–10 solve the given differential equation by using an appropriate substitution.

1. $(x - y) dx + x dy = 0$

2. $(x + y) dx + x dy = 0$

3. $x dx + (y - 2x) dy = 0$

4. $y dx = 2(x + y) dy$

5. $(y^2 + yx) dx - x^2 dy = 0$

6. $(y^2 + yx) dx + x^2 dy = 0$

7. $\frac{dy}{dx} = \frac{y - x}{y + x}$

8. $\frac{dy}{dx} = \frac{x + 3y}{3x + y}$

9. $-y dx + (x + \sqrt{xy}) dy = 0$

10. $x \frac{dy}{dx} = y + \sqrt{x^2 - y^2}, \quad x > 0$

Note that these equations are homogenous

and $y = ux$ usually is an appropriate substitution

1. $(x - y)dx + xdy = 0$

$$y = ux \rightarrow dy = udx + xdu \rightarrow (x - ux) dx + x(u dx + x du) = 0$$

$$dx + x du = 0$$

$$\frac{dx}{x} + du = 0$$

$$\ln |x| + u = c$$

$$x \ln |x| + y = cx$$

$$2. (x + y)dx + xdy = 0$$

$$y = ux \rightarrow dy = udx + xdu \rightarrow (x + ux)dx + x(udx + xdu) = 0$$

$$(1 + u)dx + udx + xdu = 0$$

$$(1 + 2u)dx + xdu = 0$$

$$\frac{du}{1 + 2u} + \frac{dx}{x} = 0$$

$$\frac{1}{2} \ln|1 + 2u| + \ln|x| = c_1$$

$$\ln|1 + 2u| + 2 \ln|x| = 2c_1 \rightarrow (1 + 2u)x^2 = c$$

$$x^2 + 2x^2 \frac{y}{x} = c$$

$$x^2 + 2xy = c$$

3. $x dx + (y - 2x) dy = 0$

$$y = ux \rightarrow dy = u dx + x du \rightarrow x dx + (ux - 2x)(u dx + x du) = 0$$

$$dx + (u - 2)(u dx + x du) = 0$$

$$(1 + u^2 - 2u) dx + x(u - 2) du = 0$$

$$\frac{(u - 2) du}{1 + u^2 - 2u} + \frac{dx}{x} = 0 \rightarrow \frac{u - 1}{(u - 1)^2} du + \frac{-1}{(u - 1)^2} du + \frac{dx}{x} = 0$$

$$\rightarrow \ln |u - 1| + \frac{1}{u - 1} + \ln |x| = c_1$$

$$\ln |(u - 1)x| + \frac{1}{u - 1} = c_1 \rightarrow \ln \left| \left(\frac{y}{x} - 1 \right) x \right| + \frac{1}{\frac{y}{x} - 1} = c_1$$

$$\rightarrow \ln |y - x| + \frac{x}{y - x} = c_1$$

$$3. \quad xdx + (y - 2x)dy = 0$$

$$x = vy \rightarrow dx = vdy + ydv \rightarrow \begin{aligned} vy(v dy + y dv) + (y - 2vy) dy &= 0 \\ vy^2 dv + y(v^2 - 2v + 1) dy &= 0 \end{aligned}$$

❖ Alternatively, we can substitute x

$$\frac{v dv}{(v - 1)^2} + \frac{dy}{y} = 0$$

$$\ln |v - 1| - \frac{1}{v - 1} + \ln |y| = c$$

$$\ln \left| \frac{x}{y} - 1 \right| - \frac{1}{x/y - 1} + \ln |y| = c$$

$$\rightarrow \ln |x - y| - \frac{y}{x - y} = c_1$$

$$9. \quad -ydx + (x + \sqrt{xy})dy = 0$$

$$y = ux \rightarrow dy = udx + xdu \rightarrow -ux dx + (x + \sqrt{u}x)(u dx + x du) = 0$$

$$(x^2 + x^2\sqrt{u}) du + xu^{3/2} dx = 0$$

$$\left(u^{-3/2} + \frac{1}{u}\right) du + \frac{dx}{x} = 0$$

$$-2u^{-1/2} + \ln |u| + \ln |x| = c$$

$$\ln |y/x| + \ln |x| = 2\sqrt{x/y} + c$$

$$y(\ln |y| - c)^2 = 4x.$$

$$10. \quad x \frac{dy}{dx} = y + \sqrt{x^2 - y^2}, \quad x > 0$$

After rearrangement

$$y = ux \rightarrow dy = udx + xdu \rightarrow \sqrt{x^2 - u^2 x^2} dx - x^2 du = 0$$

$$x\sqrt{1 - u^2} dx - x^2 du = 0, \quad (x > 0)$$

$$\frac{dx}{x} - \frac{du}{\sqrt{1 - u^2}} = 0$$

$$\ln x - \sin^{-1} u = c$$

$$\sin^{-1} u = \ln x + c_1$$

$$\sin^{-1} \frac{y}{x} = \ln x + c_2$$

$$\rightarrow y = x \sin (\ln x + c_2)$$

$$\frac{y}{x} = \sin (\ln x + c_2)$$

In Problems 11–14 solve the given initial-value problem.

11. $xy^2 \frac{dy}{dx} = y^3 - x^3, \quad y(1) = 2$

12. $(x^2 + 2y^2) \frac{dx}{dy} = xy, \quad y(-1) = 1$

13. $(x + ye^{y/x}) dx - xe^{y/x} dy = 0, \quad y(1) = 0$

14. $y dx + x(\ln x - \ln y - 1) dy = 0, \quad y(1) = e$

11. $xy^2 \frac{dy}{dx} = y^3 - x^3, \quad y(1) = 2$

$$y = ux \rightarrow dy = u dx + x du \rightarrow (x^3 - u^3 x^3) dx + u^2 x^3 (u dx + x du) = 0$$

$$dx + u^2 x du = 0$$

$$\frac{dx}{x} + u^2 du = 0$$

$$\ln |x| + \frac{1}{3} u^3 = c$$

$$3x^3 \ln |x| + y^3 = c_1 x^3$$

$$y(1) = 2 \rightarrow 3x^3 \ln |x| + y^3 = 8x^3$$

You also can replace $\frac{dy}{dx}$ as

$$\frac{dy}{dx} = x \frac{du}{dx} + u$$

13. $(x + ye^{y/x}) dx - xe^{y/x} dy = 0, \quad y(1) = 0$

$$y = ux \rightarrow dy = udx + xdu \rightarrow (x + uxe^u) dx - xe^u(u dx + x du) = 0$$

$$dx - xe^u du = 0$$

$$\frac{dx}{x} - e^u du = 0$$

$$\ln |x| - e^u = c$$

$$\ln |x| - e^{y/x} = c.$$

$$y(1) = 0 \rightarrow 0 - 1 = c \rightarrow c = -1$$

In Problems 15–20 solve the given differential equation by using an appropriate substitution.

15. $x \frac{dy}{dx} + y = \frac{1}{y^2}$

16. $\frac{dy}{dx} - y = e^x y^2$

17. $\frac{dy}{dx} = y(xy^3 - 1)$

18. $x \frac{dy}{dx} - (1 + x)y = xy^2$

19. $t^2 \frac{dy}{dt} + y^2 = ty$

20. $3(1 + t^2) \frac{dy}{dt} = 2ty(y^3 - 1)$

$$15. \quad x \frac{dy}{dx} + y = \frac{1}{y^2} \quad \rightarrow \quad \frac{dy}{dx} + \frac{1}{x}y = \frac{1}{x}y^{-2}$$

$$w = y^3 \rightarrow dw = 3y^2 dy \rightarrow \quad \frac{dw}{dx} + \frac{3}{x}w = \frac{3}{x}$$

$$P(x) = \frac{3}{x} \rightarrow e^{\int \frac{3}{x} dx} = x^3 \rightarrow x^3 \frac{dw}{dx} + x^2 w = 3x^2$$

$$\rightarrow \frac{d}{dx}[x^3 w] = 3x^2$$

$$\rightarrow x^3 w = x^3 + c \rightarrow y^3 = 1 + cx^{-3}$$

$$\frac{dy}{dx} + P(x)y = f(x)y^n,$$

Bernoulli's equation

$$w = y^{1-n} \quad \rightarrow dw = (1-n)y^{-n}dy \rightarrow \dots$$

$$16. \quad \frac{dy}{dx} - y = e^x y^2 \quad \rightarrow y^{-2} \frac{dy}{dx} - y^{-1} = e^x$$

$$w = y^{1-2} = y^{-1} \rightarrow \frac{dw}{dx} = -1y^{-2} \frac{dy}{dx} \rightarrow -\frac{dw}{dx} - w = e^x \rightarrow \frac{dw}{dx} + w = -e^x$$

$$P(x) = +1 \rightarrow e^{\int dx} = e^x \rightarrow e^x \frac{dw}{dx} + e^x w = -e^{2x} \rightarrow \frac{d}{dx} [e^x w] = -e^{2x}$$

$$\rightarrow e^x w = -\frac{1}{2} e^{2x} + c \rightarrow w = -\frac{1}{2} e^x + c e^{-x}$$

$$\rightarrow y^{-1} = -\frac{1}{2} e^x + c e^{-x}$$

$$17. \quad \frac{dy}{dx} = y(xy^3 - 1) \quad \rightarrow \quad \frac{dy}{dx} + y = xy^4 \quad \rightarrow \quad y^{-4} \frac{dy}{dx} + y^{-3} = x$$

$$w = y^{1-4} = y^{-3} \quad \rightarrow \quad \frac{dw}{dx} = -3y^{-4} \frac{dy}{dx} \quad \rightarrow \quad \frac{dw}{dx} - 3w = -3x$$

$$P(x) = -3 \quad \rightarrow \quad e^{-3 \int dx} = e^{-3x} \quad \rightarrow \quad e^{-3x} \frac{dw}{dx} - 3e^{-3x} w = -3xe^{-3x}$$

$$\rightarrow \frac{d}{dx} [e^{-3x} w] = -3xe^{-3x}$$

$$\rightarrow e^{-3x} w = xe^{-3x} + \frac{1}{3} e^{-3x} + c \quad \rightarrow \quad w = x + \frac{1}{3} e^x + ce^{-3x}$$

$$\rightarrow y^{-3} = x + \frac{1}{3} e^x + ce^{-3x}$$

In Problems 21 and 22 solve the given initial-value problem.

21. $x^2 \frac{dy}{dx} - 2xy = 3y^4, \quad y(1) = \frac{1}{2}$

22. $y^{1/2} \frac{dy}{dx} + y^{3/2} = 1, \quad y(0) = 4$

$$21. \quad x^2 \frac{dy}{dx} - 2xy = 3y^4, \quad y(1) = \frac{1}{2} \quad \rightarrow \quad \frac{dy}{dx} - \frac{2}{x}y = \frac{3}{x^2}y^4 \quad \rightarrow \quad y^{-4} \frac{dy}{dx} - \frac{2}{x}y^{-3} = \frac{3}{x^2}$$

$$w = y^{1-4} = y^{-3} \quad \rightarrow \quad \frac{dw}{dx} = -3y^{-4} \frac{dy}{dx} \quad \rightarrow \quad \frac{dw}{dx} + \frac{6}{x}w = \frac{-9}{x^2}$$

$$P(x) = \frac{6}{x} \quad \rightarrow \quad e^{\int \frac{6}{x} dx} = x^6 \quad \rightarrow \quad x^6 \frac{dw}{dx} + 6x^5 w = -9x^4$$

$$\rightarrow \frac{d}{dx} [x^6 w] = -9x^4$$

$$\rightarrow x^6 w = -\frac{9}{5}x^5 + c \quad \rightarrow \quad w = -\frac{9}{5}x^{-1} + cx^{-6}$$

$$\rightarrow y^{-3} = -\frac{9}{5}x^{-1} + cx^{-6}$$

$$y(1) = \frac{1}{2} \rightarrow 8 = -\frac{9}{5} + c \rightarrow c = -\frac{49}{5} \quad \rightarrow \quad y^{-3} = -\frac{9}{5}x^{-1} - \frac{49}{5}x^{-6}$$

$$22. \quad y^{1/2} \frac{dy}{dx} + y^{3/2} = 1, \quad y(0) = 4 \quad \rightarrow \quad \frac{dy}{dx} + y = y^{\frac{-1}{2}} \quad \rightarrow \quad y^{\frac{1}{2}} \frac{dy}{dx} + y^{\frac{3}{2}} = 1$$

$$w = y^{1-\frac{-1}{2}} = y^{\frac{3}{2}} \quad \rightarrow \quad \frac{dw}{dx} = \frac{3}{2} y^{\frac{1}{2}} \frac{dy}{dx} \quad \rightarrow \quad \frac{dw}{dx} + \frac{3}{2} w = \frac{3}{2}$$

$$P(x) = \frac{3}{2} \quad \rightarrow \quad e^{\int \frac{3}{2} dx} = e^{\frac{3}{2}x} \quad \rightarrow \quad e^{\frac{3}{2}x} \frac{dw}{dx} + \frac{3}{2} e^{\frac{3}{2}x} w = \frac{3}{2} e^{\frac{3}{2}x}$$

$$\rightarrow \frac{d}{dx} [e^{\frac{3}{2}x} w] = \frac{3}{2} e^{\frac{3}{2}x}$$

$$\rightarrow e^{\frac{3}{2}x} w = e^{\frac{3}{2}x} + c \quad \rightarrow \quad w = 1 + c e^{-\frac{3}{2}x}$$

$$\rightarrow y^{\frac{3}{2}} = 1 + c e^{-\frac{3}{2}x}$$

$$y(0) = 4 \rightarrow 8 = 1 + c \rightarrow c = 7 \rightarrow y^{-3} = 1 + 7 e^{-\frac{3}{2}x}$$

In Problems 23–28 solve the given differential equation by using an appropriate substitution.

23. $\frac{dy}{dx} = (x + y + 1)^2$

24. $\frac{dy}{dx} = \frac{1 - x - y}{x + y}$

25. $\frac{dy}{dx} = \tan^2(x + y)$

26. $\frac{dy}{dx} = \sin(x + y)$

27. $\frac{dy}{dx} = 2 + \sqrt{y - 2x + 3}$

28. $\frac{dy}{dx} = 1 + e^{y-x+5}$

23. $\frac{dy}{dx} = (x + y + 1)^2$



$u = x + y + 1$

$$u = x + y + 1 \rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$$

$$\frac{du}{dx} - 1 = u^2 \rightarrow \frac{1}{1+u^2} du = dx$$

$$\tan^{-1} u = x + c$$

$$u = \tan(x + c)$$

$$x + y + 1 = \tan(x + c)$$

$$y = \tan(x + c) - x - 1$$

24. $\frac{dy}{dx} = \frac{1 - x - y}{x + y}$



$u = x + y \rightarrow$

$$u = x + y \rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$$

$$\frac{du}{dx} - 1 = \frac{1-u}{u} \rightarrow u \, du = dx$$

$$\frac{1}{2}u^2 = x + c$$

$$u^2 = 2x + c_1$$

$$(x + y)^2 = 2x + c_1$$

In Problems 29 and 30 solve the given initial-value problem.

29. $\frac{dy}{dx} = \cos(x + y), \quad y(0) = \pi/4$

30. $\frac{dy}{dx} = \frac{3x + 2y}{3x + 2y + 2}, \quad y(-1) = -1$

29. $\frac{dy}{dx} = \cos(x + y), \quad y(0) = \pi/4$



$$u = x + y$$

$$u = x + y \rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$$

$$\frac{du}{dx} - 1 = \cos u \rightarrow \frac{1}{1 + \cos u} du = dx$$

$$\frac{1}{1 + \cos u} = \frac{1 - \cos u}{1 - \cos^2 u} = \frac{1 - \cos u}{\sin^2 u} = \csc^2 u - \csc u \cot u$$

You may use the table 

$$\int (\csc^2 u - \csc u \cot u) du = \int dx$$

$$-\cot u + \csc u = x + c$$

$$-\cot(x + y) + \csc(x + y) = x + c$$

$$x = 0, y = \frac{\pi}{4} \rightarrow c = \sqrt{2} - 1$$

$$\csc(x + y) - \cot(x + y) = x + \sqrt{2} - 1$$

$$30. \quad \frac{dy}{dx} = \frac{3x + 2y}{3x + 2y + 2}, \quad y(-1) = -1$$



$$u = 3x + 2y$$

$$u = 3x + 2y \rightarrow \frac{du}{dx} = 3 + 2\frac{dy}{dx}$$

$$\frac{du}{dx} = 3 + \frac{2u}{u+2} = \frac{5u+6}{u+2} \rightarrow \frac{u+2}{5u+6} du = dx$$

$$\frac{u+2}{5u+6} = \frac{1}{5} + \frac{4}{25u+30}$$

$$\int \left(\frac{1}{5} + \frac{4}{25u+30} \right) du = \int dx$$

$$\frac{1}{5}u + \frac{4}{25} \ln |25u + 30| = x + c$$

$$\frac{1}{5}(3x + 2y) + \frac{4}{25} \ln |75x + 50y + 30| = x + c$$

$$x = -1, y = -1 \rightarrow c = \frac{4}{25} \ln 95$$

Replace c and simplify it $\longrightarrow$

$$5y - 5x + 2 \ln |75x + 50y + 30| = 2 \ln 95$$

Basic

Integral

Formulas

$$1. \int u^n du = \frac{u^{n+1}}{n+1} + C \text{ (for } n \neq -1)$$

$$2. \int \frac{1}{u} du = \ln |u| + C$$

$$3. \int du = u + C$$

$$4. \int e^u du = e^u + C$$

$$5. \int a^u du = \frac{a^u}{\ln a} + C \text{ (for } a > 0, a \neq 1)$$

$$6. \int \ln u du = u \ln u - u + C$$

$$7. \int \sin u du = -\cos u + C$$

$$8. \int \cos u du = \sin u + C$$

$$9. \int \sin^2 u du = \frac{u}{2} - \frac{\sin 2u}{4} + C$$

$$10. \int \cos^2 u du = \frac{u}{2} + \frac{\sin 2u}{4} + C$$

$$11. \int \tan u du = -\ln |\cos u| + C = \ln |\sec u| + C$$

$$12. \int \sec u du = \ln |\sec u + \tan u| + C$$

$$13. \int \sec^2 u du = \tan u + C$$

$$14. \int \sec u \tan u du = \sec u + C$$

$$15. \int \cot u du = \ln |\sin u| + C = -\ln |\csc u| + C$$

$$16. \int \csc u du = \ln |\csc u - \cot u| + C$$